

Chapter 1

Fundamental Concepts

Signals

- A **signal** is a **pattern of variation of a physical quantity**, often as a *function* of time (but also space, distance, position, etc).
- These quantities are usually the **independent variables** of the function defining the signal
- A signal encodes **information**, which is the variation itself

Signal Processing

- Signal processing is the discipline concerned with **extracting, analyzing, and manipulating the information** carried by signals
- The processing method depends on the type of signal and on the nature of the information carried by the signal

Characterization and Classification of Signals

- The **type of signal** depends on the nature of the independent variables and on the value of the function defining the signal
- For example, the independent variables can be **continuous or discrete**
- Likewise, the signal can be a **continuous or discrete function** of the independent variables

Characterization and Classification of Signals – Cont'd

- Moreover, the signal can be either a **real-valued function** or a **complex-valued function**
- A signal consisting of a single component is called a **scalar or one-dimensional (1-D) signal**

Examples: CT vs. DT Signals

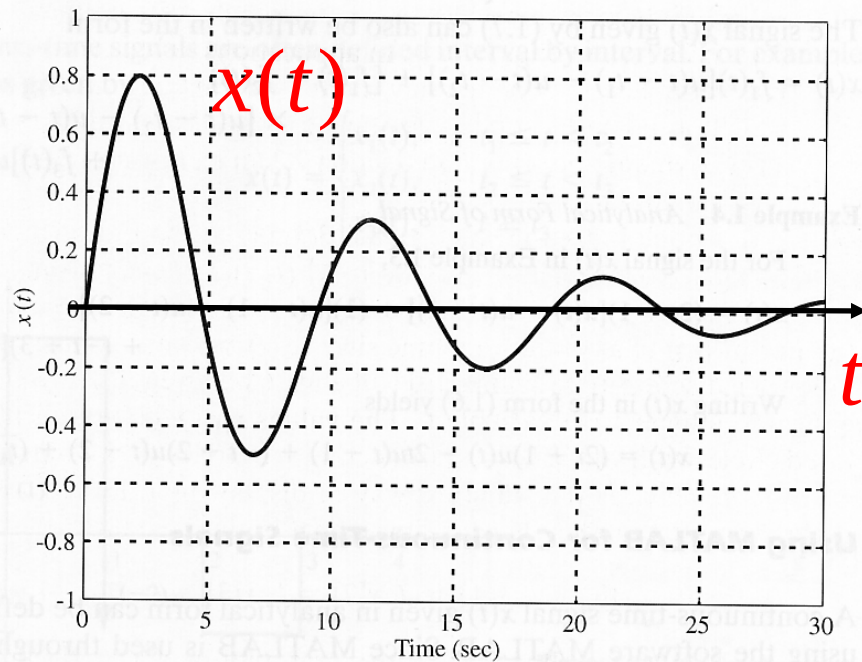


Figure 1.13 MATLAB plot of the signal $x(t) = e^{-0.1t} \sin \frac{\pi}{3} t$.

`plot(t,x)`

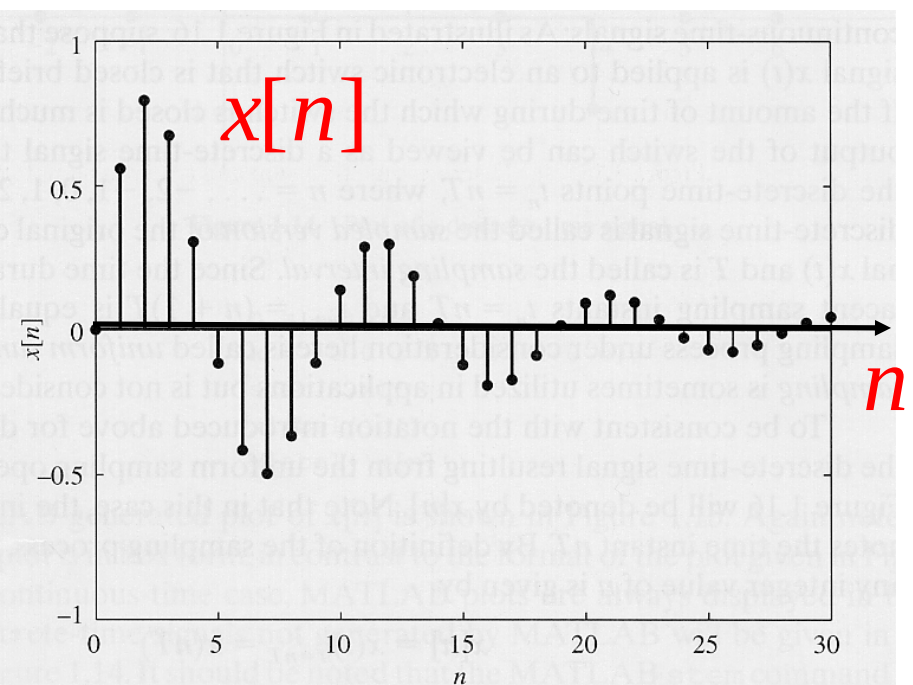


Figure 1.17 Sampled continuous-time signal.

`stem(n,x)`

Sampling

- Discrete-time signals are often obtained by sampling continuous-time signals

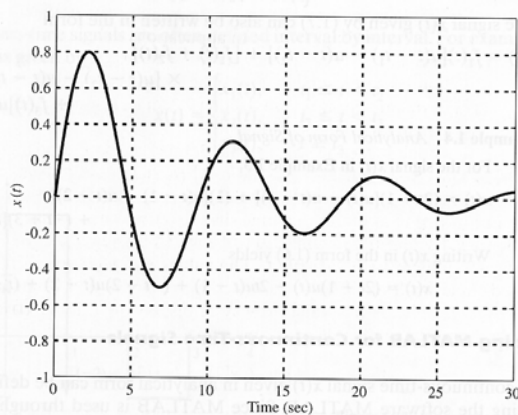
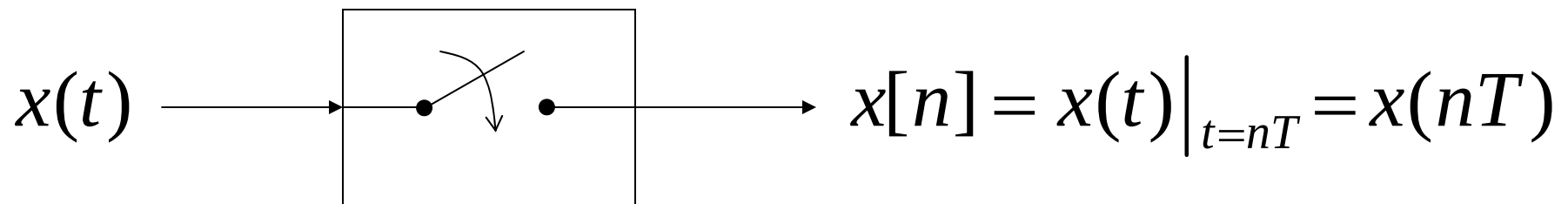


Figure 1.13 MATLAB plot of the signal $x(t) = e^{-0.1t} \sin \frac{3}{8} t$.

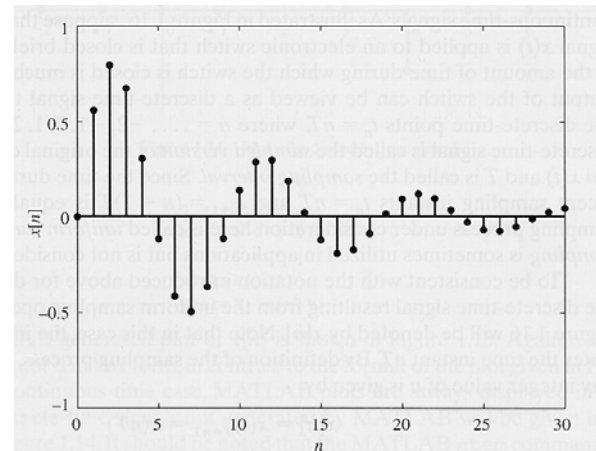


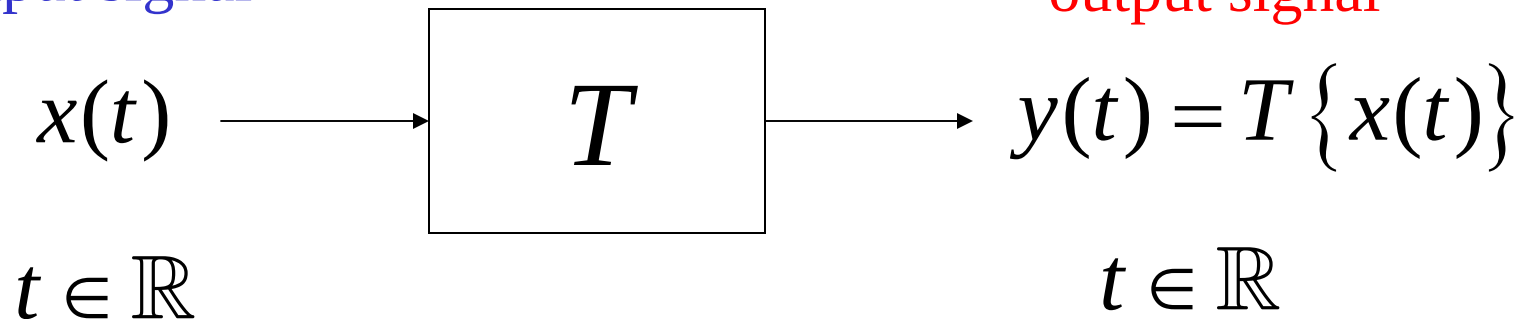
Figure 1.17 Sampled continuous-time signal.

Systems

- A **system** is any device that can **process signals** for analysis, synthesis, enhancement, format conversion, recording, transmission, etc.
- A system is usually mathematically defined by the equation(s) relating input to output signals (**I/O characterization**)
- A system may have single or multiple inputs and single or multiple outputs

Block Diagram Representation of Single-Input Single-Output (SISO) CT Systems

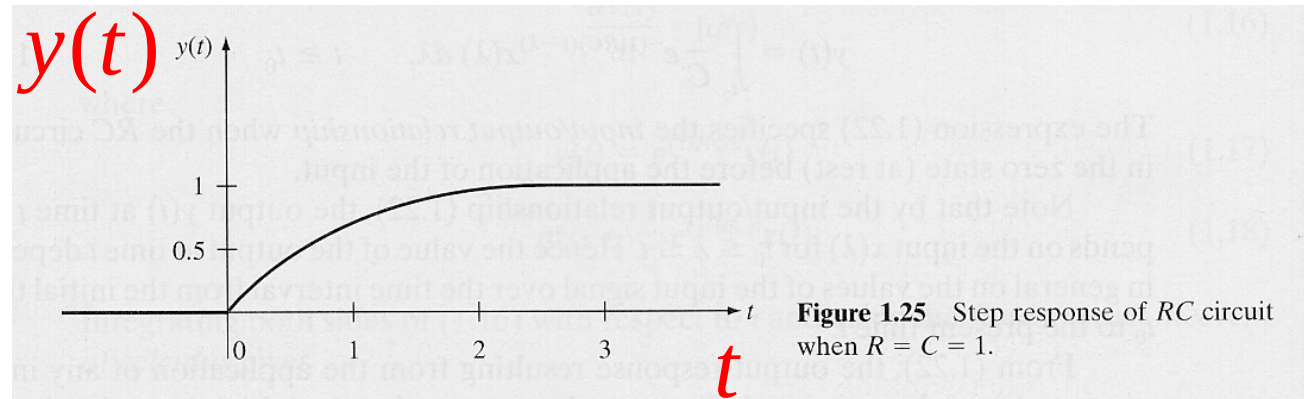
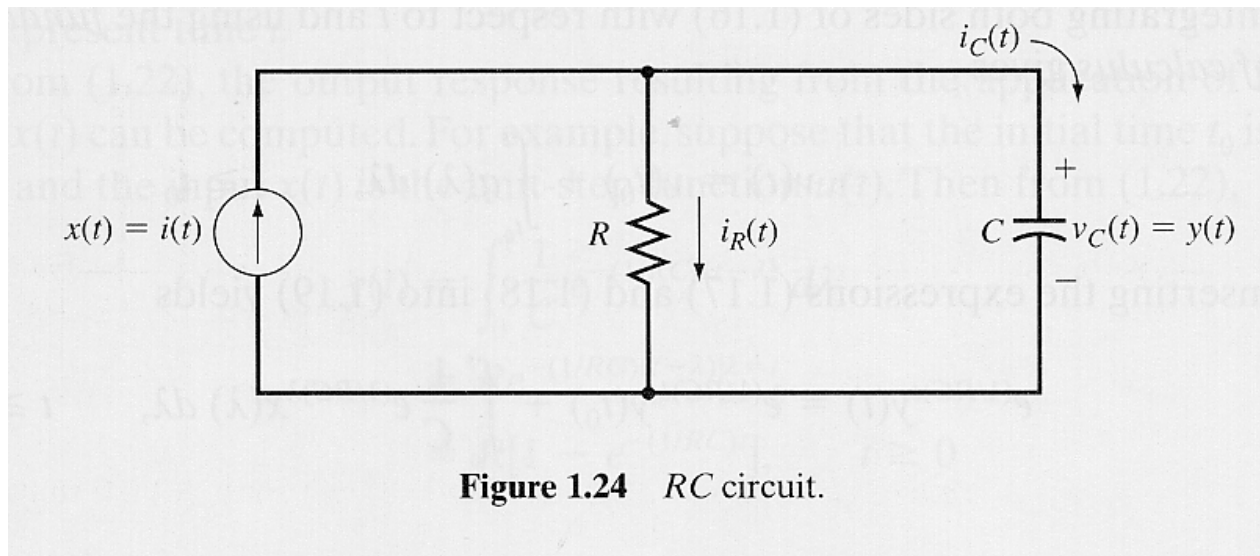
input signal



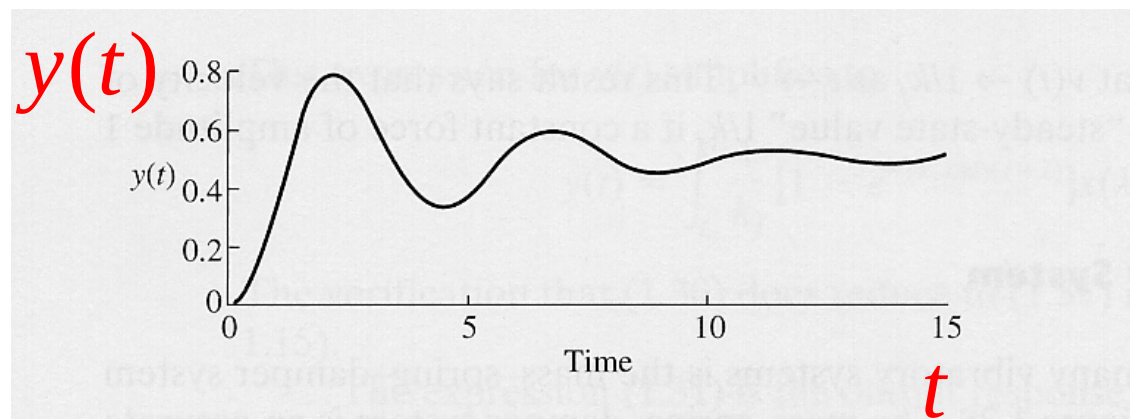
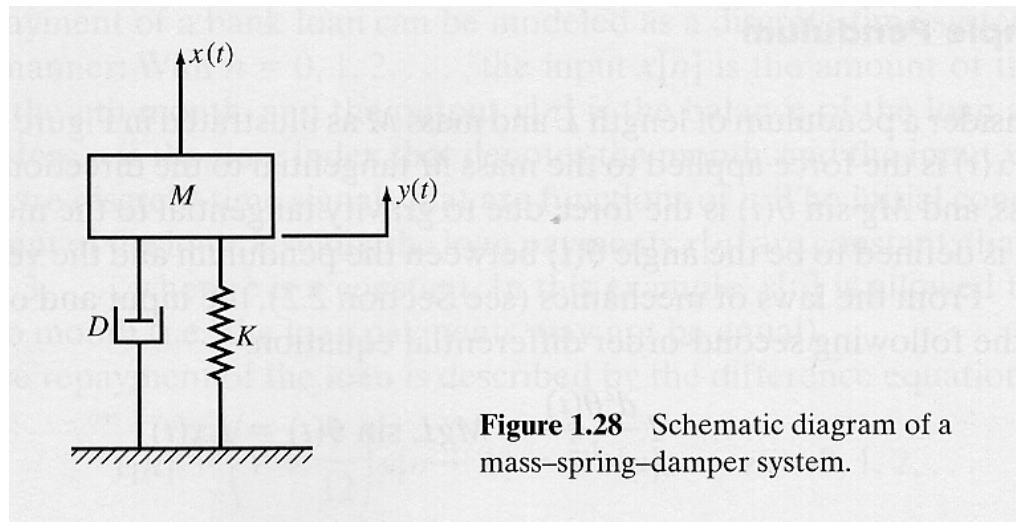
Types of input/output representations considered

- Differential equation
- Convolution model
- Transfer function representation (Fourier transform, Laplace transform)

Examples of 1-D, Real-Valued, CT Signals: Temporal Evolution of Currents and Voltages in Electrical Circuits

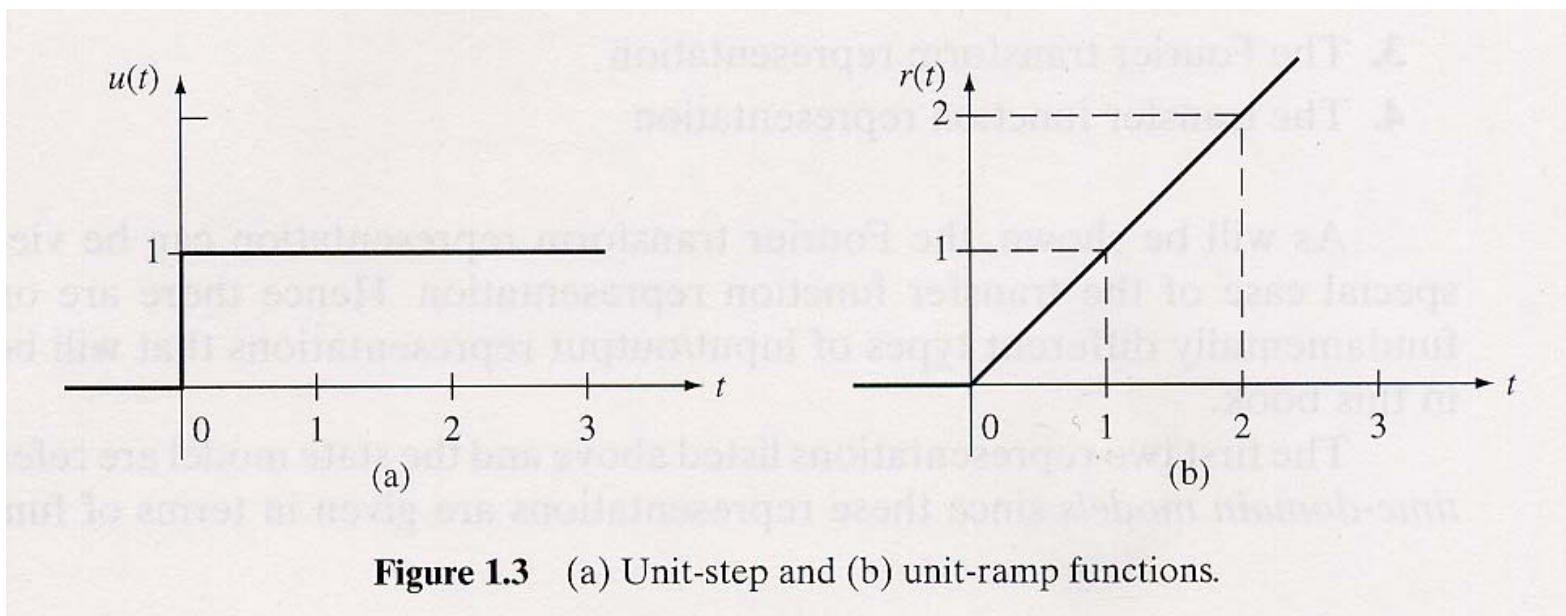


Examples of 1-D, Real-Valued, CT Signals: Temporal Evolution of Some Physical Quantities in Mechanical Systems



Continuous-Time (CT) Signals

- Unit-step function $u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$
- Unit-ramp function $r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$



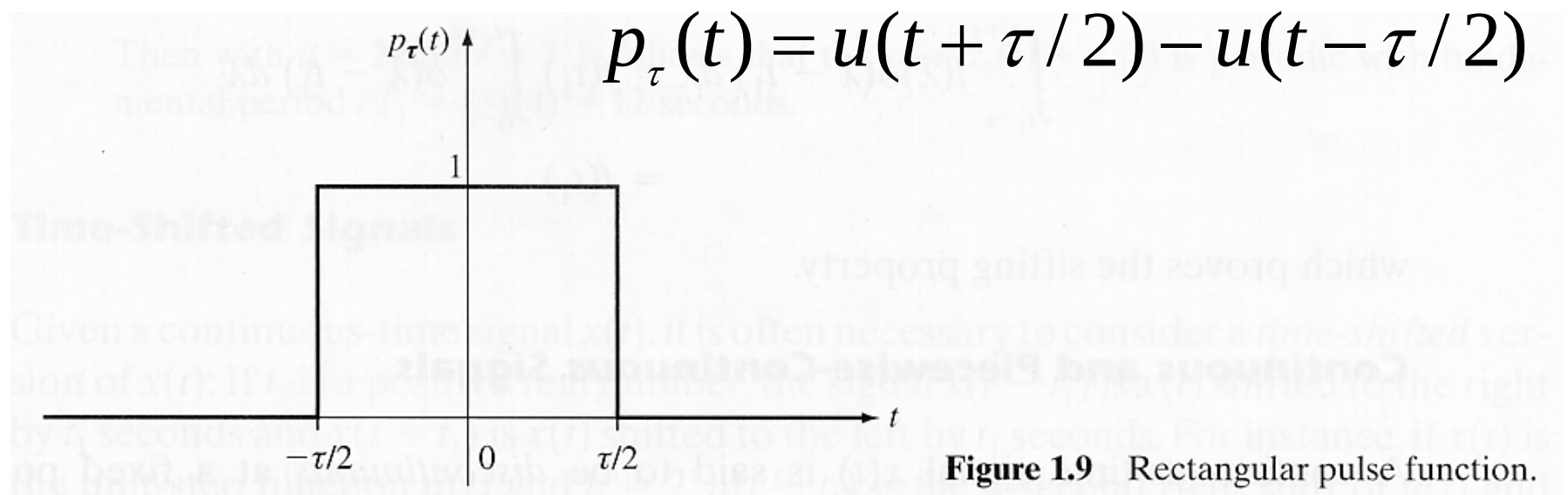
Unit-Ramp and Unit-Step Functions: Some Properties

$$x(t)u(t) = \begin{cases} x(t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$r(t) = \int_{-\infty}^t u(\lambda) d\lambda$$

$$u(t) = \frac{dr(t)}{dt} \quad (\text{with exception of } t = 0)$$

The Rectangular Pulse Function



The Unit Impulse

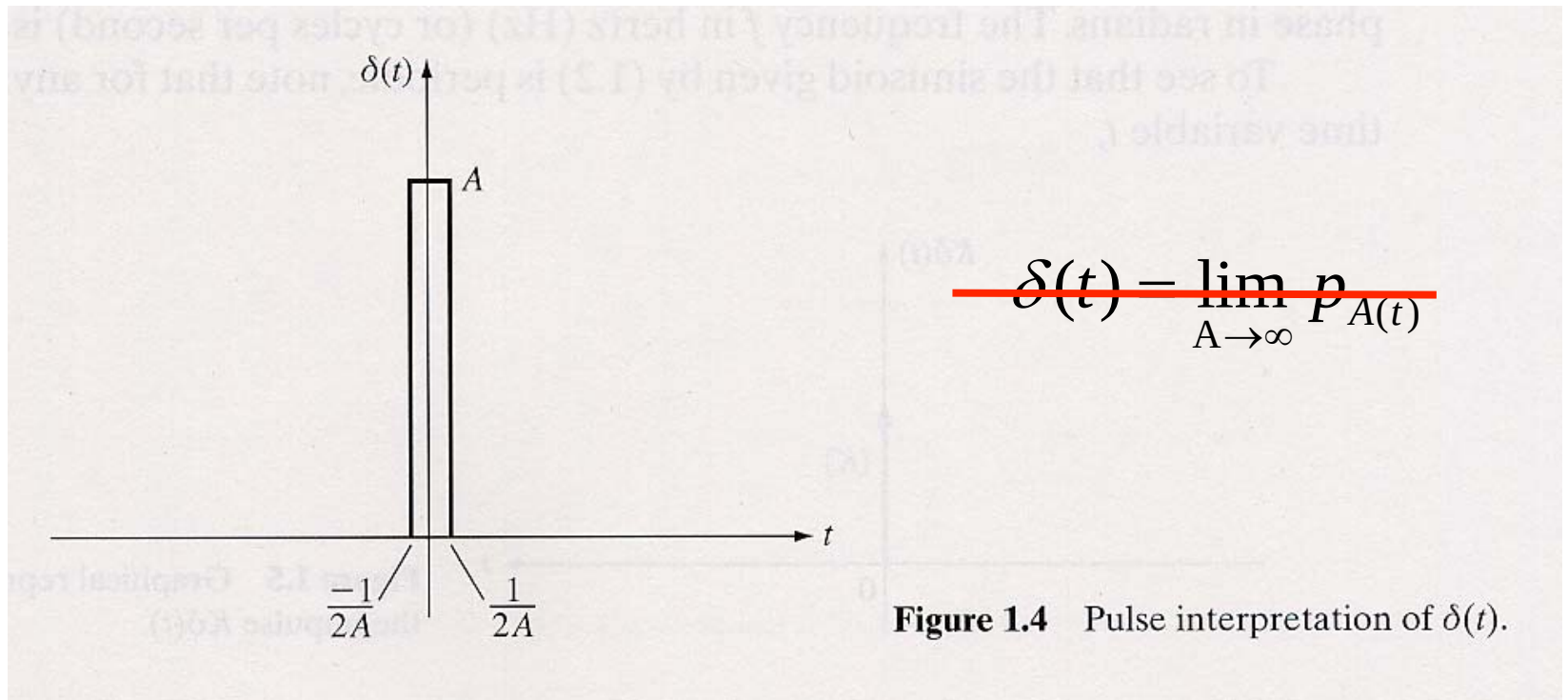
- A.k.a. the **delta function** or **Dirac distribution**
- It is defined by:

$$\delta(t) = 0, \quad t \neq 0$$

$$\int_{-\varepsilon}^{\varepsilon} \delta(\lambda) d\lambda = 1, \quad \forall \varepsilon > 0$$

- The value $\delta(0)$ is not defined, in particular $\delta(0) \neq \infty$

The Unit Impulse: Graphical Interpretation



A is a very large number

The Scaled Impulse $K\delta(t)$

- If $K \in \mathbb{R}$, $K\delta(t)$ is the impulse with area K , i.e.,

$$K\delta(t) = 0, \quad t \neq 0$$

$$\int_{-\varepsilon}^{\varepsilon} K\delta(\lambda) d\lambda = K, \quad \forall \varepsilon > 0$$

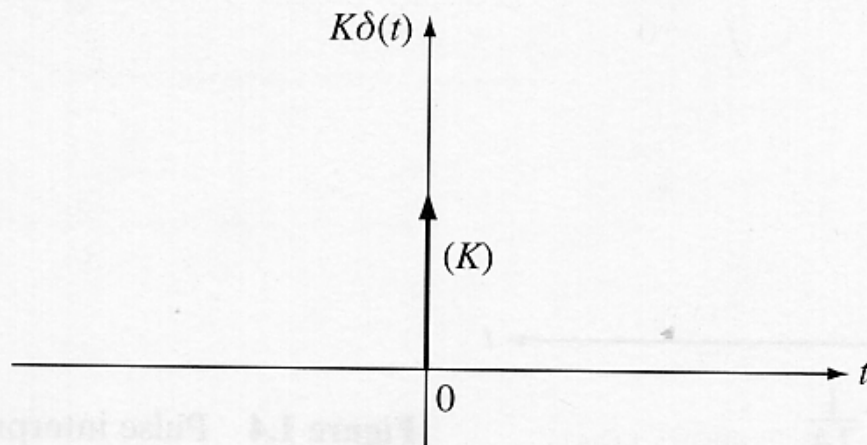


Figure 1.5 Graphical representation of the impulse $K\delta(t)$.

Properties of the Delta Function

$$1) \quad u(t) = \int_{-\infty}^t \delta(\lambda) d\lambda$$

$\forall t$ except $t = 0$

$$2) \quad \int_{t_0 - \varepsilon}^{t_0 + \varepsilon} x(t) \delta(t - t_0) dt = x(t_0) \quad \forall \varepsilon > 0$$

(sifting property)

Periodic Signals

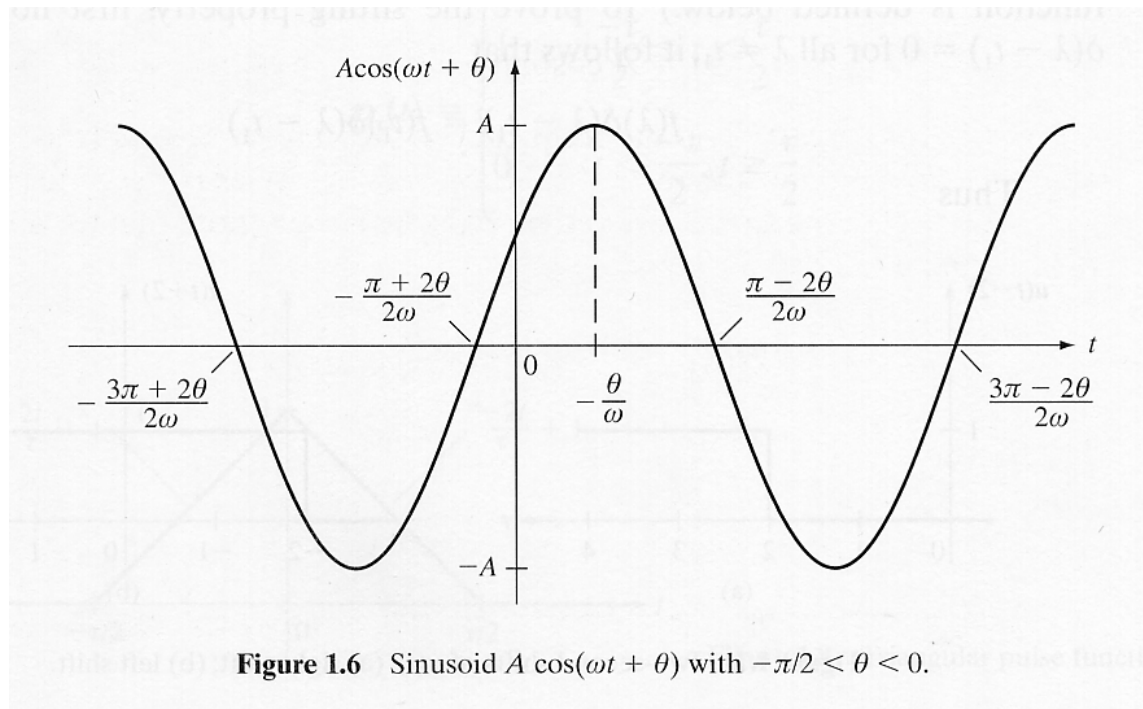
- Definition: a signal $x(t)$ is said to be periodic with period T , if

$$x(t + T) = x(t) \quad \forall t \in \mathbb{R}$$

- Notice that $x(t)$ is also periodic with period qT where q is any positive integer
- T is called the **fundamental period**

Example: The Sinusoid

$$x(t) = A \cos(\omega t + \theta), \quad t \in \mathbb{R}$$



$$\omega \text{ [rad / sec]}$$

$$\theta \text{ [rad]}$$

$$f = \frac{\omega}{2\pi} \text{ [1 / sec]} = \text{[Hz]}$$

Time-Shifted Signals

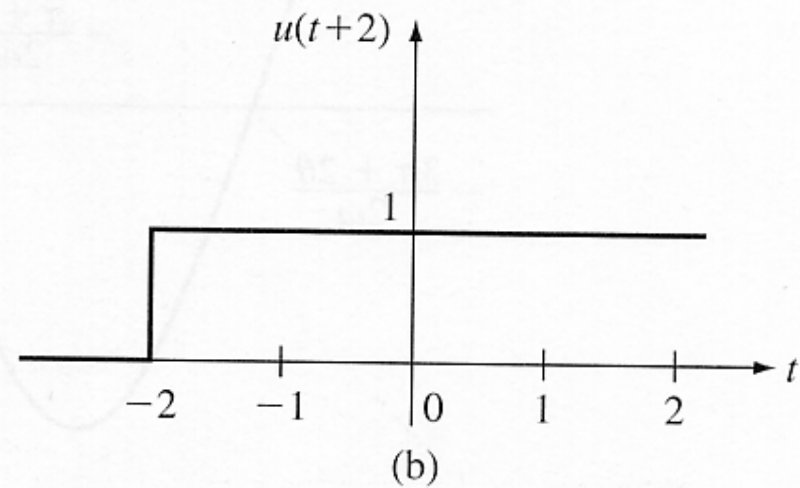
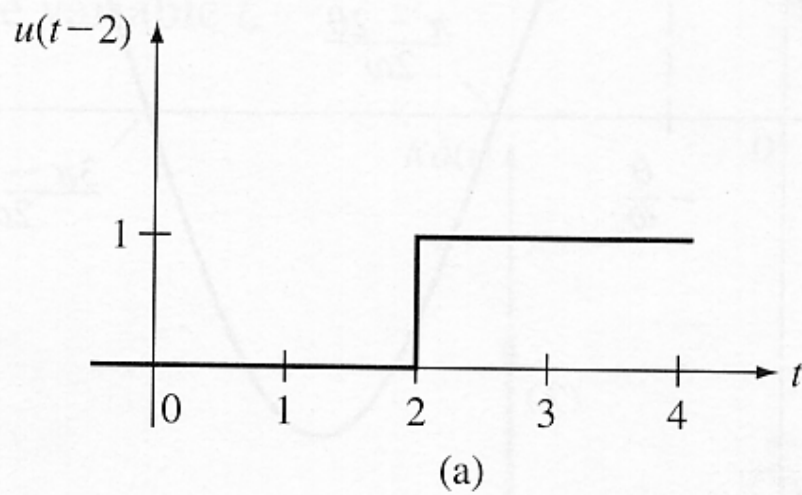
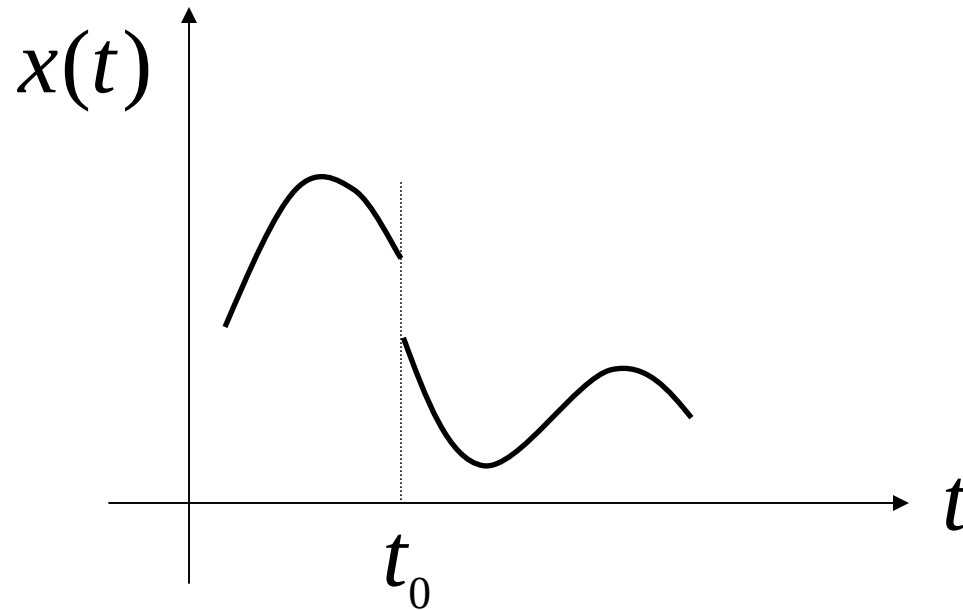


Figure 1.7 Two-second shifts of $u(t)$: (a) right shift; (b) left shift.

Points of Discontinuity

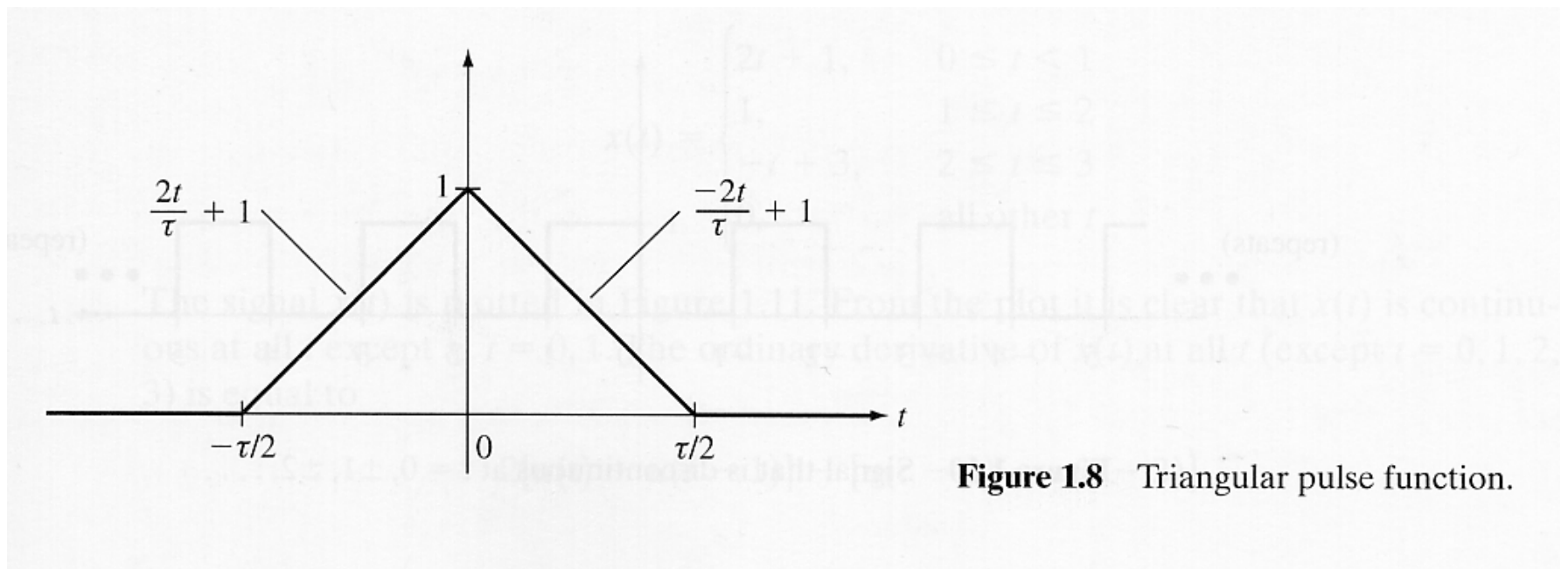
- A continuous-time signal $x(t)$ is said to be discontinuous at a point t_0 if $x(t_0^+) \neq x(t_0^-)$ where $t_0^+ = t_0 + \varepsilon$ and $t_0^- = t_0 - \varepsilon$, ε being a small positive number



Continuous Signals

- A signal $x(t)$ is continuous at the point t_0 if $x(t_0^+) = x(t_0^-)$
- If a signal $x(t)$ is continuous at all points t , $x(t)$ is said to be a **continuous signal**

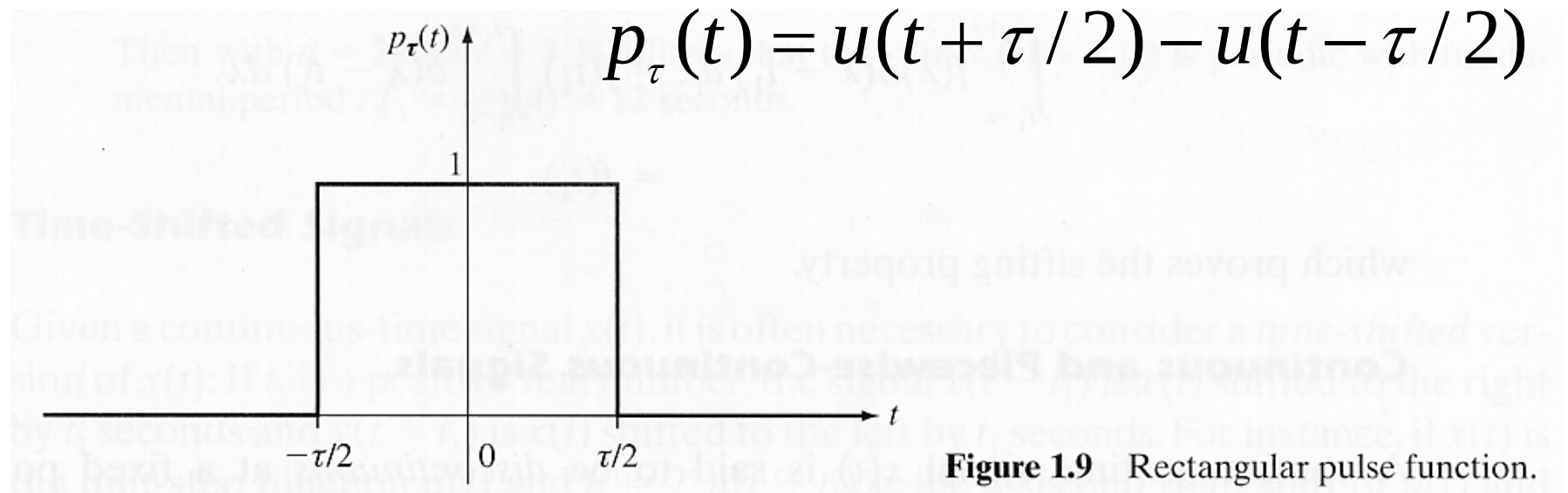
Example of Continuous Signal: The Triangular Pulse Function



Piecewise-Continuous Signals

- A signal $x(t)$ is said to be piecewise continuous if it is continuous at all t except a finite or countably infinite collection of points $t_i, i = 1, 2, 3, \dots$

Example of Piecewise-Continuous Signal: The Rectangular Pulse Function



Another Example of Piecewise-Continuous Signal: The Pulse Train Function

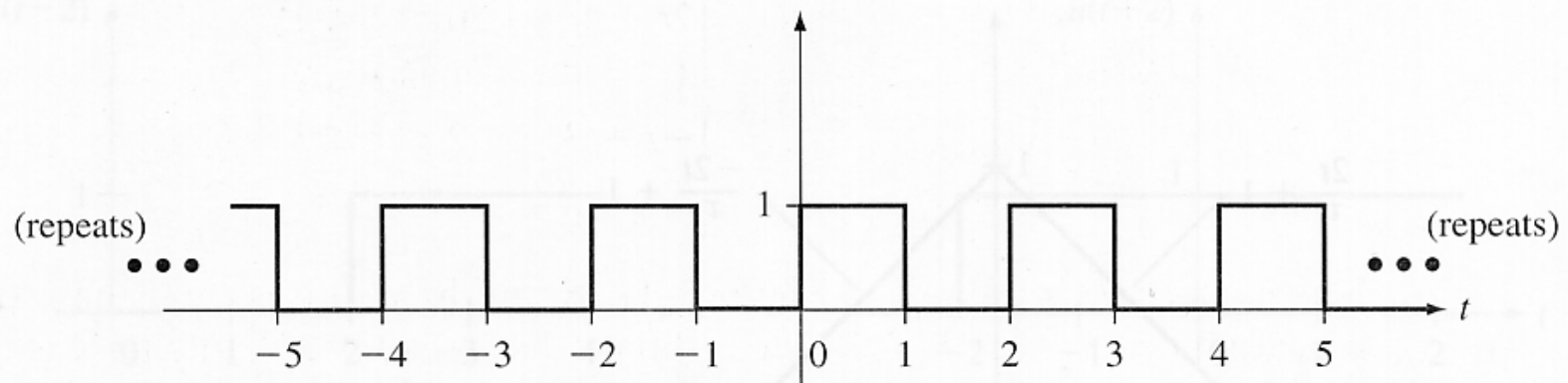


Figure 1.10 Signal that is discontinuous at $t = 0, \pm 1, \pm 2, \dots$

Derivative of a Continuous-Time Signal

- A signal $x(t)$ is said to be **differentiable** at a point t_0 if the quantity

$$\frac{x(t_0 + h) - x(t_0)}{h}$$

has limit as $h \rightarrow 0$ independent of whether h approaches 0 from above ($h > 0$) or from below ($h < 0$)

- If the limit exists, $x(t)$ has a **derivative** at t_0

$$\left. \frac{dx(t)}{dt} \right|_{t=t_0} = \lim_{h \rightarrow 0} \frac{x(t_0 + h) - x(t_0)}{h}$$

Generalized Derivative

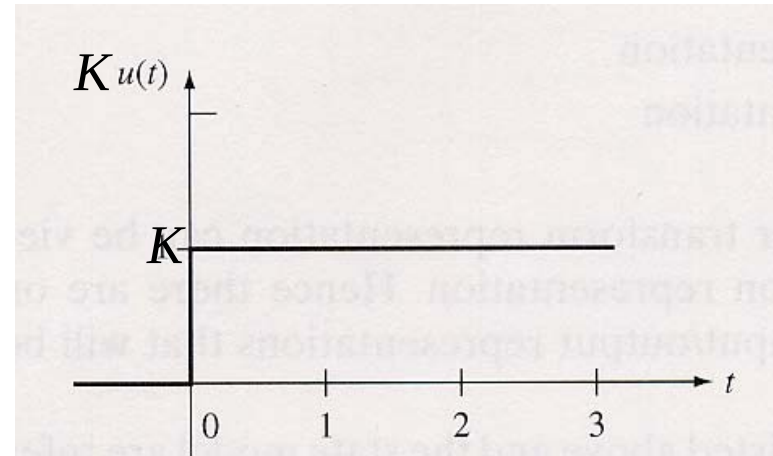
- However, piecewise-continuous signals may have a derivative in a generalized sense
- Suppose that $x(t)$ is differentiable at all t except $t = t_0$
- The **generalized derivative** of $x(t)$ is defined to be

$$\frac{dx(t)}{dt} + [x(t_0^+) - x(t_0^-)]\delta(t - t_0)$$

 ordinary derivative of $x(t)$ at all t except $t = t_0$

Example: Generalized Derivative of the Step Function

- Define $x(t) = Ku(t)$



- The ordinary derivative of $x(t)$ is 0 at all points except $t = 0$
- Therefore, the generalized derivative of $x(t)$ is

$$K[u(0^+) - u(0^-)]\delta(t - 0) = K\delta(t)$$

Another Example of Generalized Derivative

- Consider the function defined as

$$x(t) = \begin{cases} 2t + 1, & 0 \leq t < 1 \\ 1, & 1 \leq t < 2 \\ -t + 3, & 2 \leq t \leq 3 \\ 0, & \text{all other } t \end{cases}$$

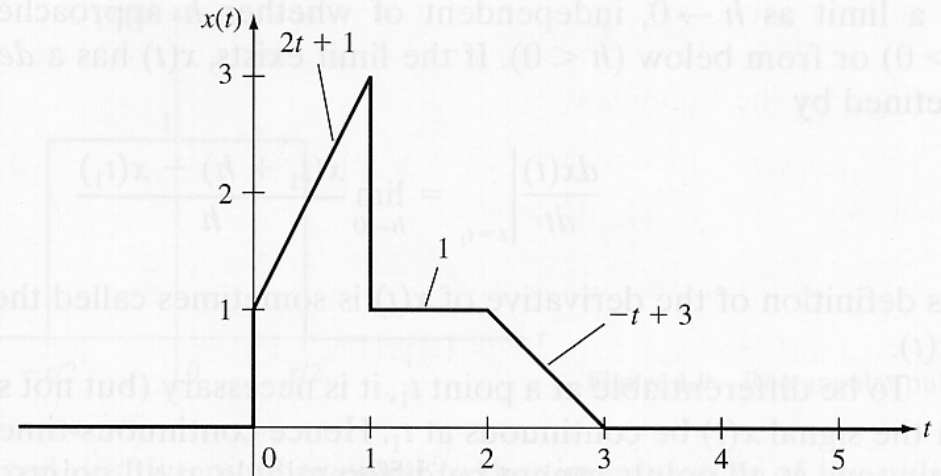


Figure 1.11 Signal in Example 1.3.

Another Example of Generalized Derivative: Cont'd

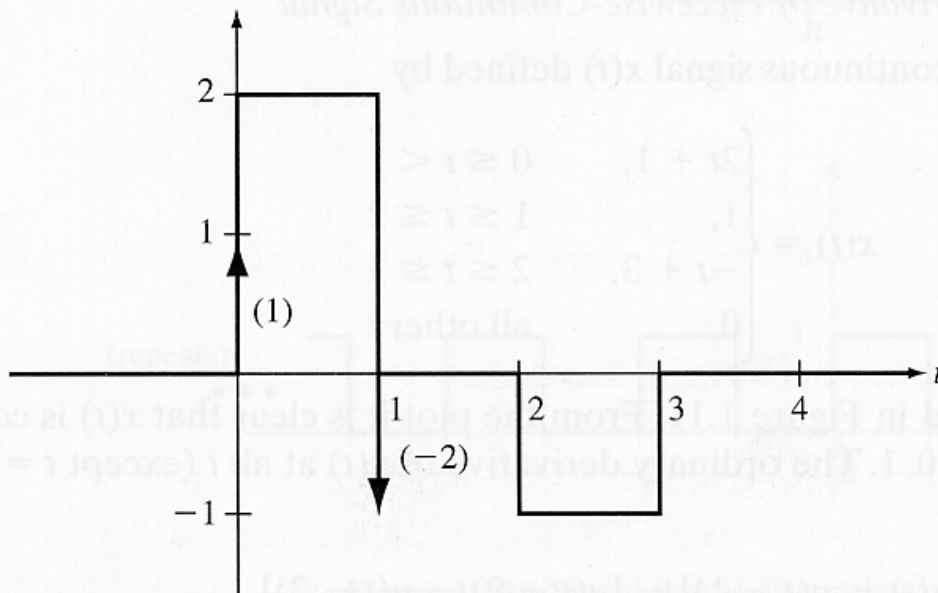


Figure 1.12 Generalized derivative of the signal in Example 1.3.

Example of CT System: An RC Circuit

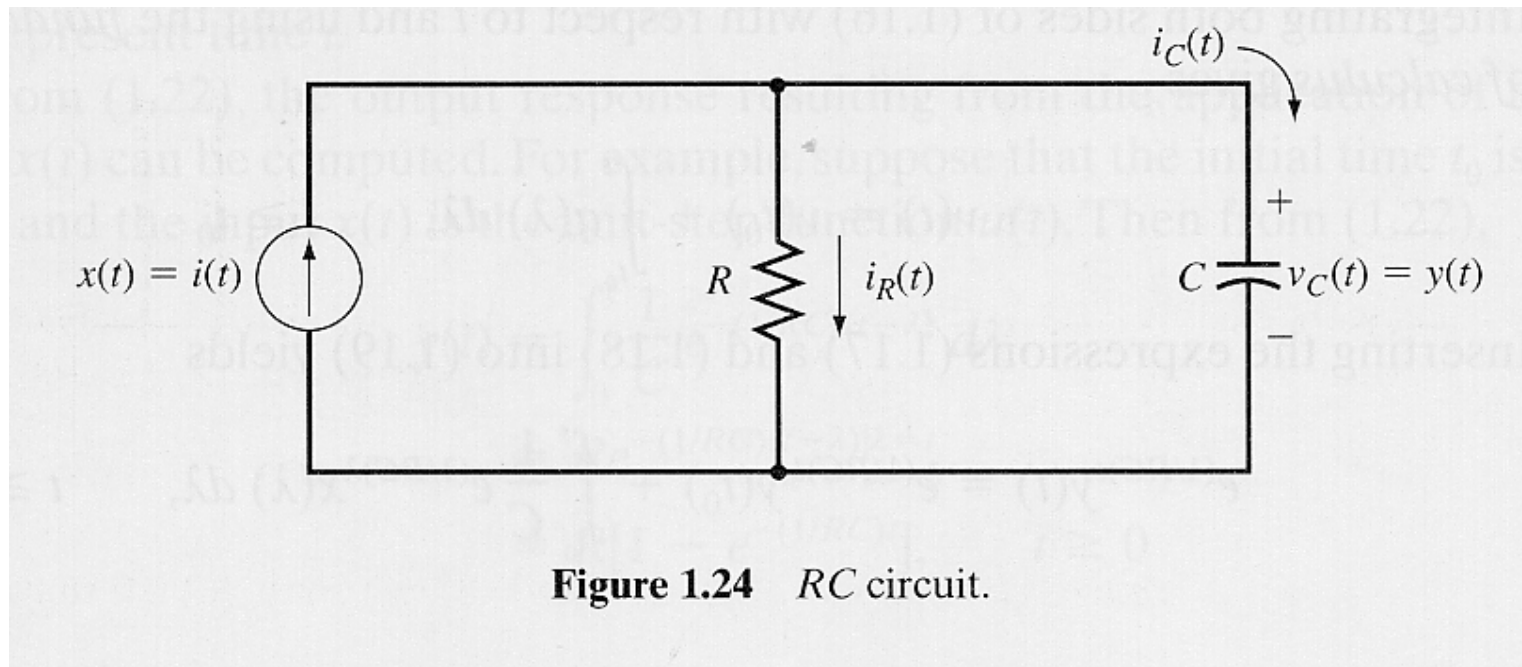


Figure 1.24 RC circuit.

Kirchhoff's current law: $i_C(t) + i_R(t) = i(t)$

RC Circuit: Cont'd

- The v - i law for the capacitor is

$$i_C(t) = C \frac{dv_C(t)}{dt} = C \frac{dy(t)}{dt}$$

- Whereas for the resistor it is

$$i_R(t) = \frac{1}{R} v_C(t) = \frac{1}{R} y(t)$$

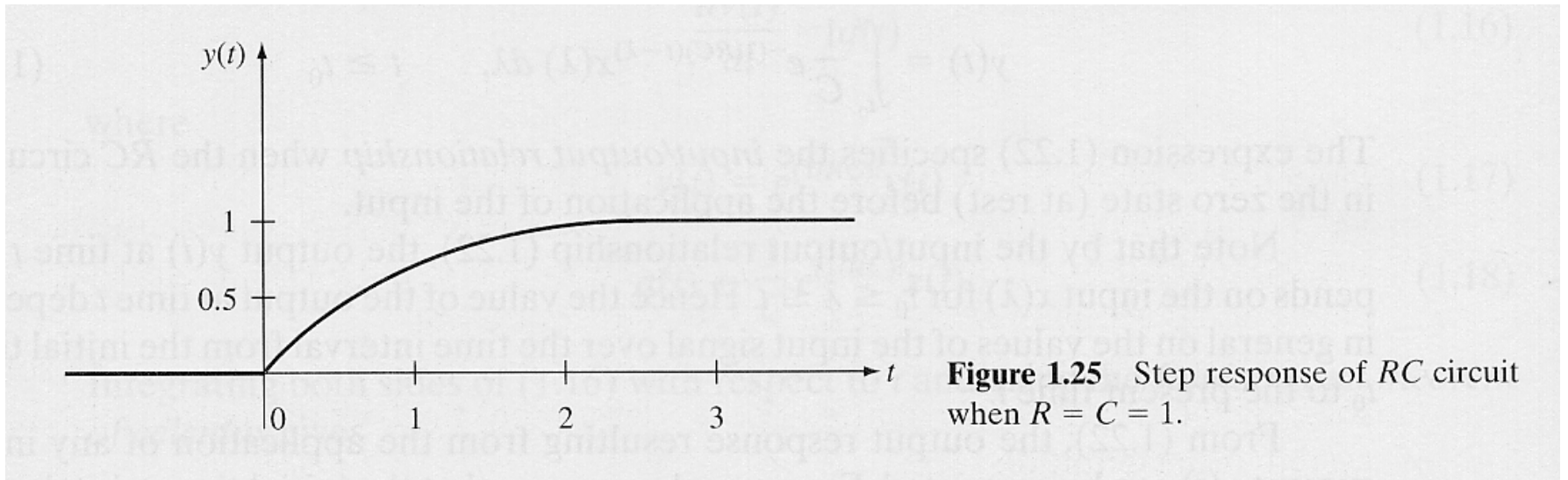
RC Circuit: Cont'd

- Constant-coefficient linear differential equation describing the I/O relationship if the circuit

$$C \frac{dy(t)}{dt} + \frac{1}{R} y(t) = i(t) = x(t)$$

RC Circuit: Cont'd

- Step response when $R=C=1$



Basic System Properties: Causality

- A system is said to be **causal** if, for any time t_1 , the output response at time t_1 resulting from input $x(t)$ does not depend on values of the input for $t > t_1$.
- A system is said to be **noncausal** if it is not causal

Example: The Ideal Predictor

$$y(t) = x(t + 1)$$

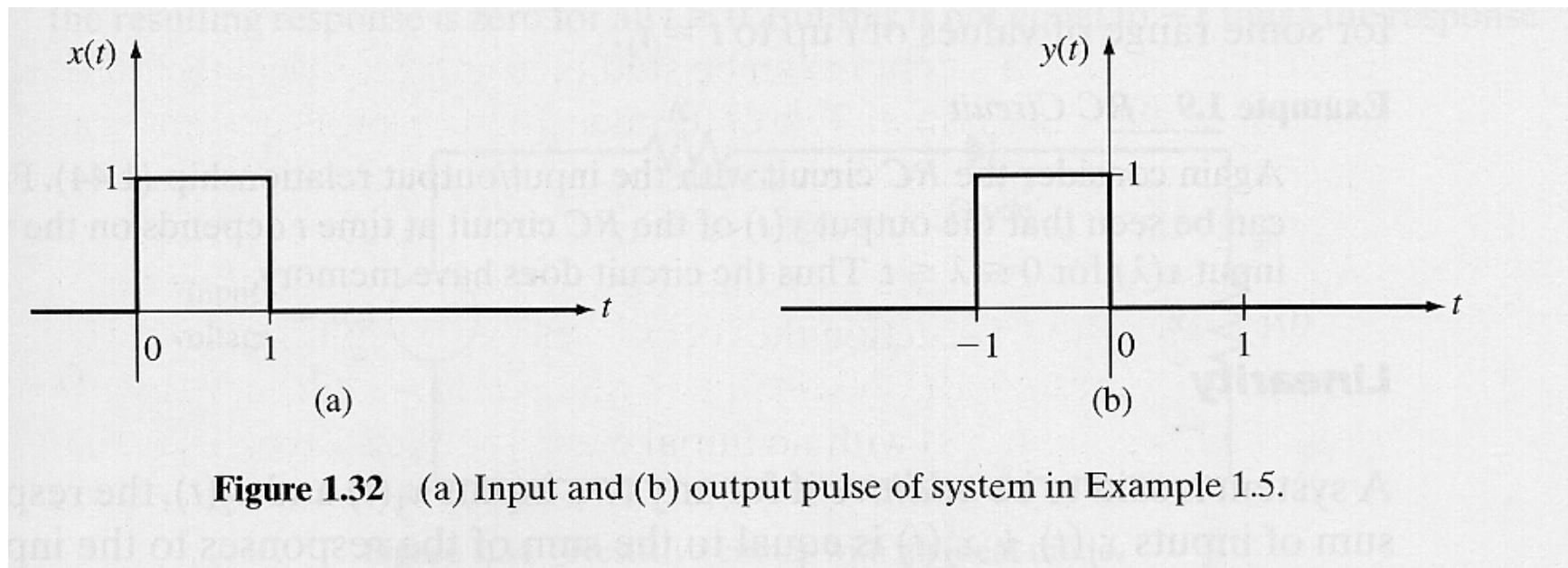
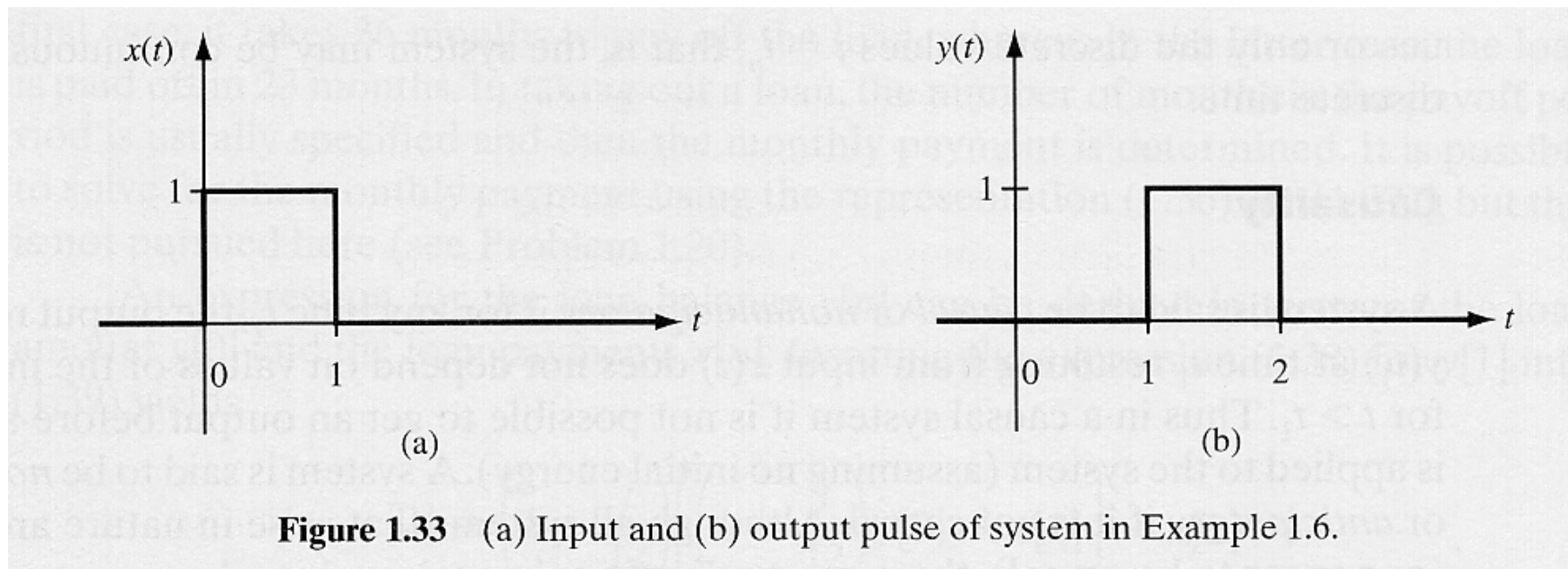


Figure 1.32 (a) Input and (b) output pulse of system in Example 1.5.

Example: The Ideal Delay

$$y(t) = x(t - 1)$$



Memoryless Systems and Systems with Memory

- A causal system is **memoryless** or **static** if, for any time t_1 , the value of the output at time t_1 depends only on the value of the input at time t_1
- A causal system that is not memoryless is said to have **memory**. A system has memory if the output at time t_1 depends in general on the past values of the input $x(t)$ for some range of values of t up to $t = t_1$

Examples

- Ideal Amplifier/Attenuator

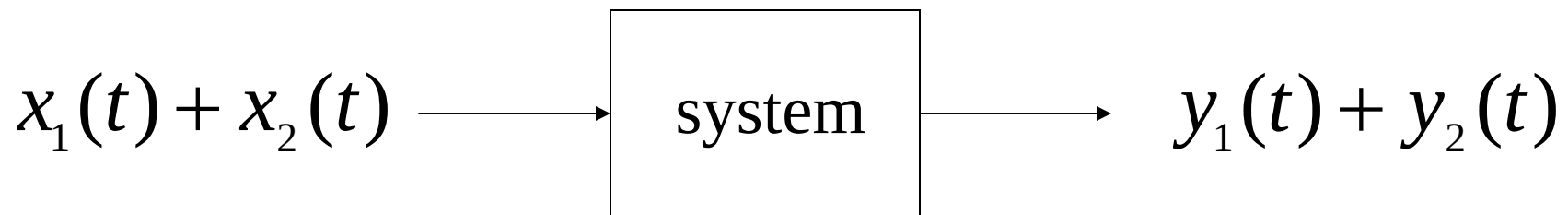
$$y(t) = Kx(t)$$

- RC Circuit

$$y(t) = \frac{1}{C} \int_0^t e^{-(1/RC)(t-\tau)} x(\tau) d\tau, \quad t \geq 0$$

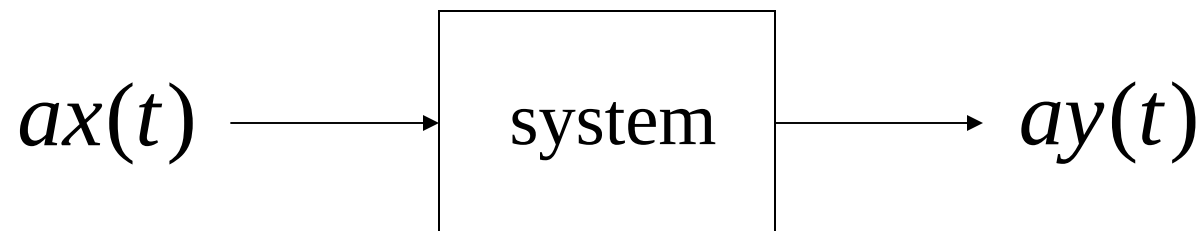
Basic System Properties: Additive Systems

- A system is said to be **additive** if, for any two inputs $x_1(t)$ and $x_2(t)$, the response to the sum of inputs $x_1(t) + x_2(t)$ is equal to the sum of the responses to the inputs (assuming no initial energy before the application of the inputs)



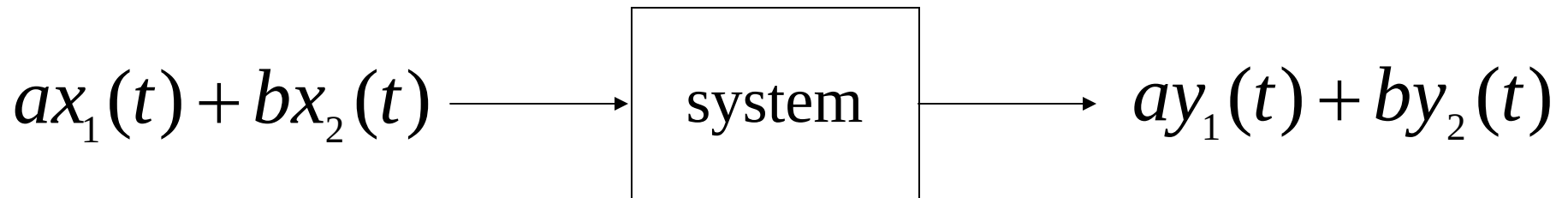
Basic System Properties: Homogeneous Systems

- A system is said to be **homogeneous** if, for any input $x(t)$ and any scalar a , the response to the input $ax(t)$ is equal to a times the response to $x(t)$, assuming no energy before the application of the input



Basic System Properties: Linearity

- A system is said to be **linear** if it is both additive and homogeneous



- A system that is not linear is said to be **nonlinear**

Example of Nonlinear System: Circuit with a Diode

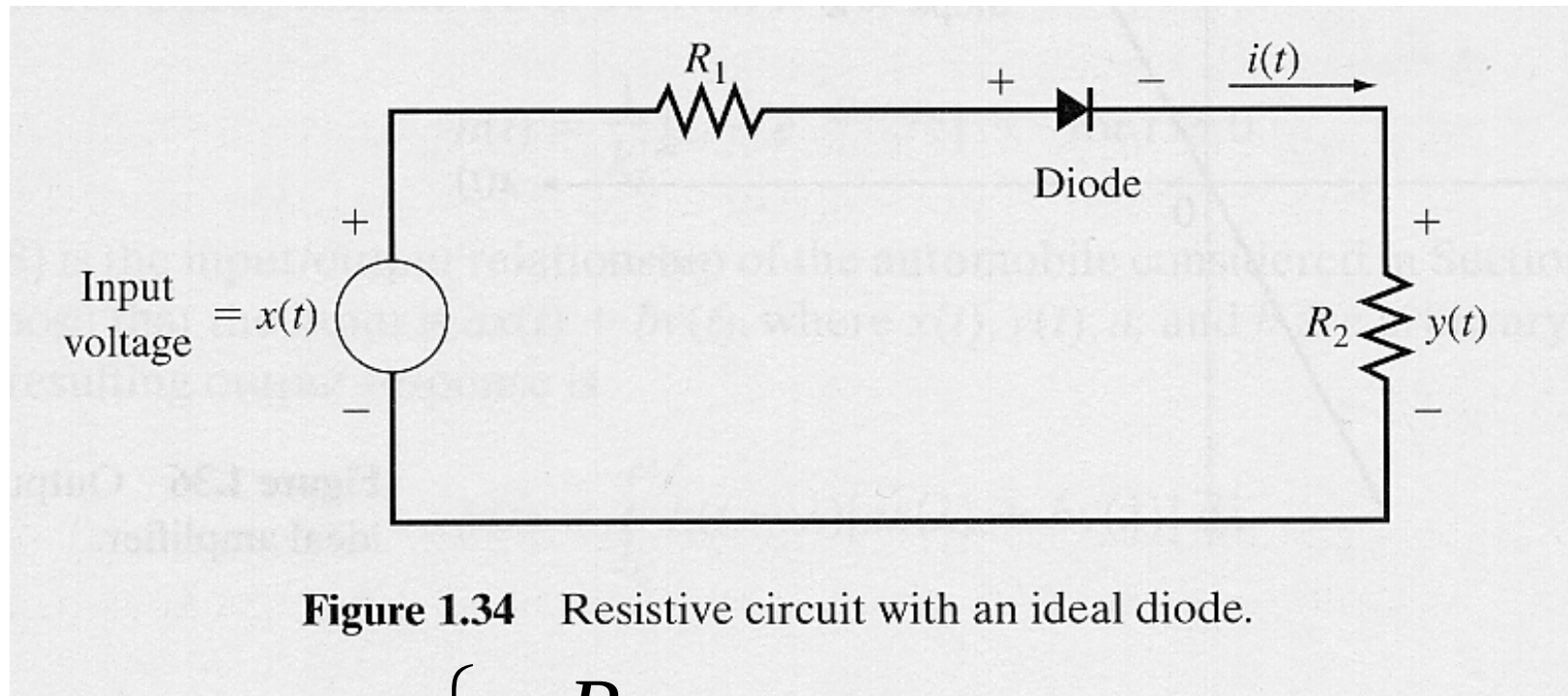
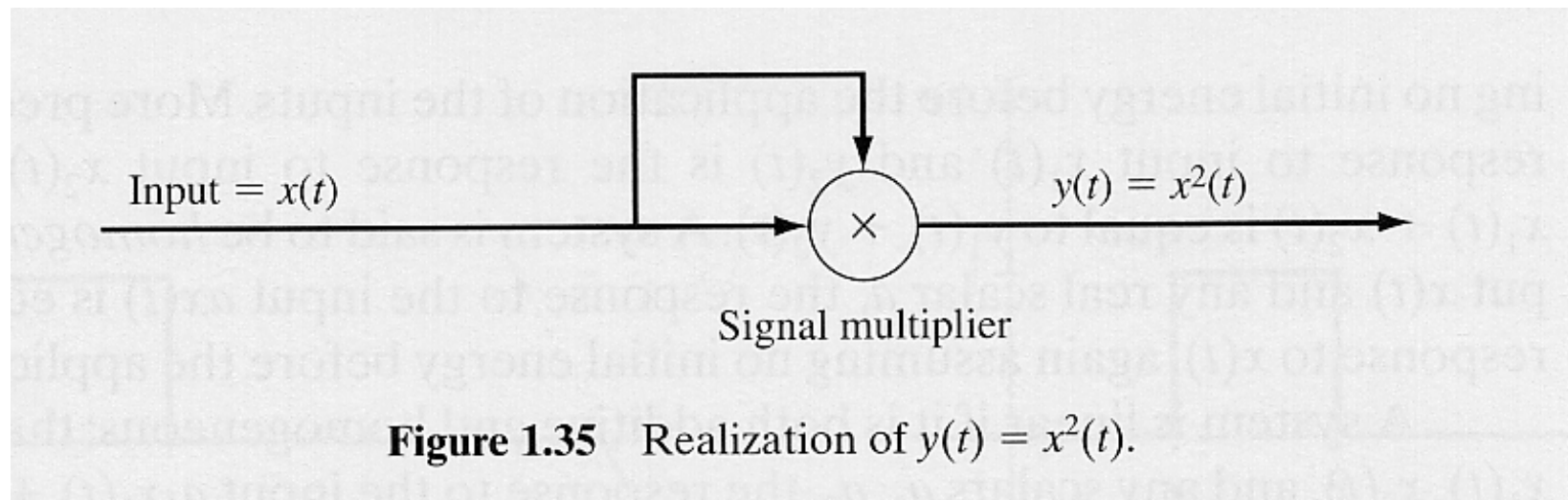


Figure 1.34 Resistive circuit with an ideal diode.

$$y(t) = \begin{cases} \frac{R_2}{R_1 + R_2} x(t), & \text{when } x(t) \geq 0 \\ 0, & \text{when } x(t) \leq 0 \end{cases}$$

Example of Nonlinear System: Square-Law Device

$$y(t) = x^2(t)$$



Example of Linear System: The Ideal Amplifier

$$y(t) = Kx(t)$$

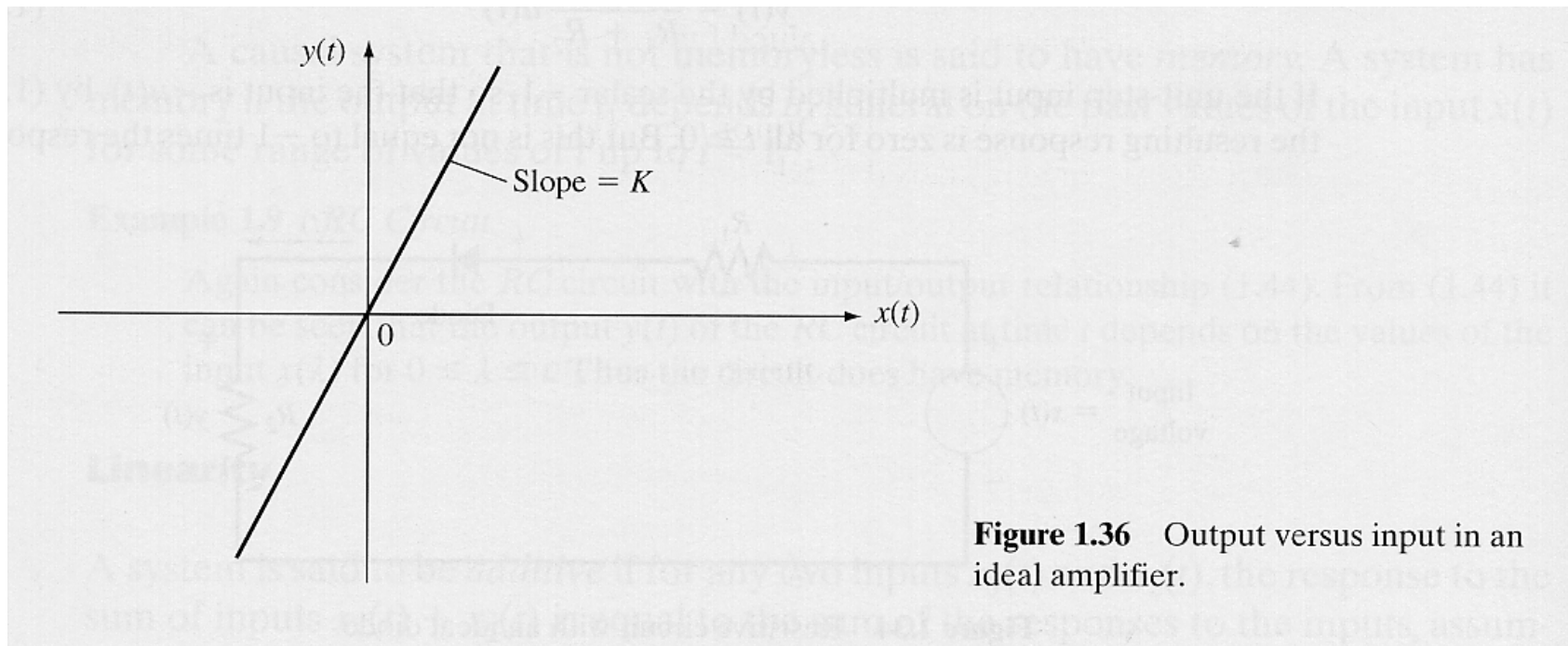
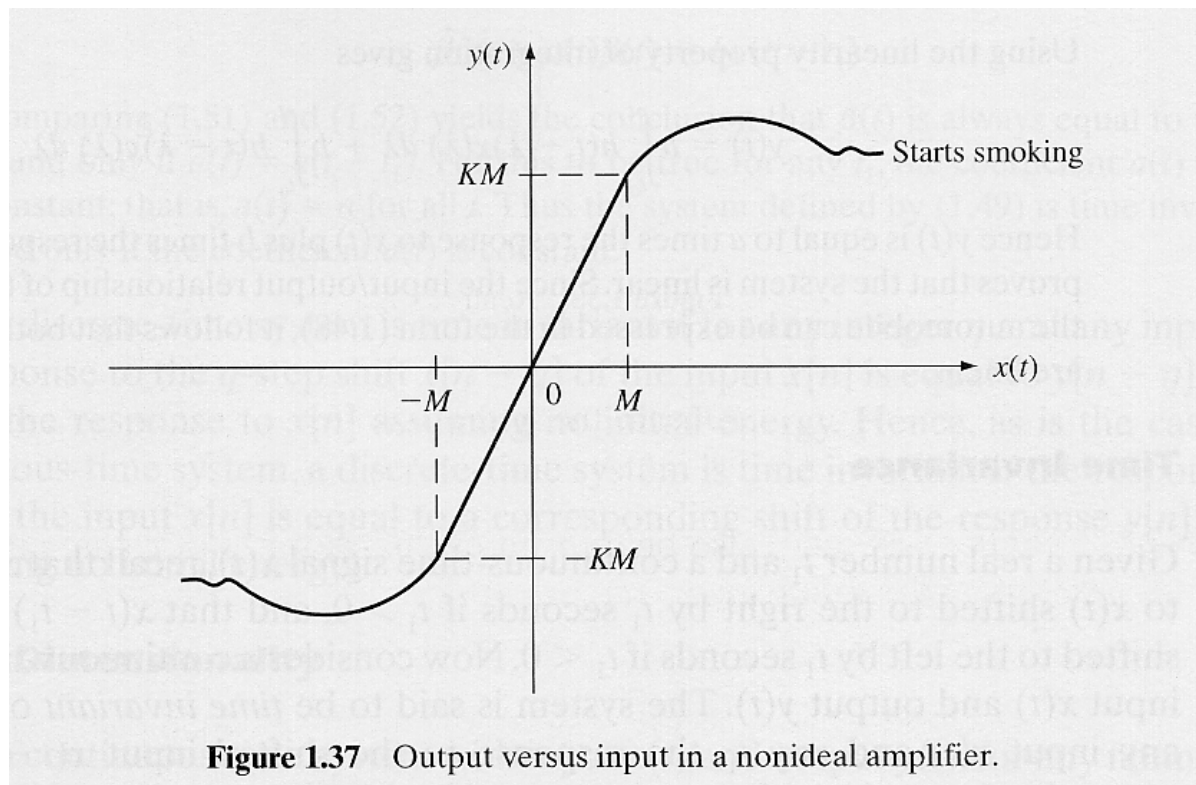


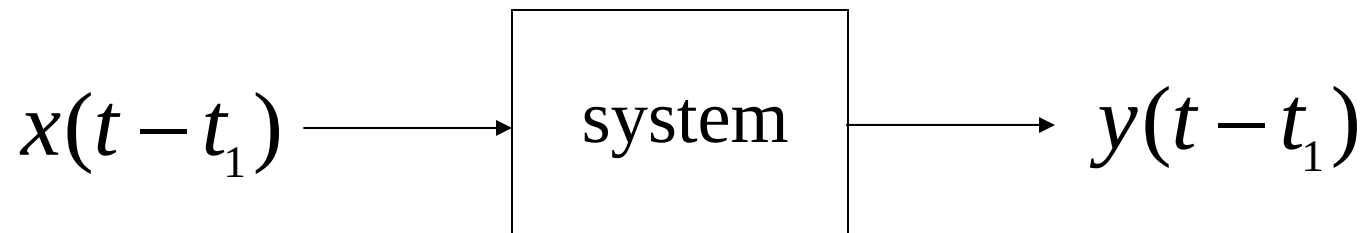
Figure 1.36 Output versus input in an ideal amplifier.

Example of Nonlinear System: A Real Amplifier



Basic System Properties: Time Invariance

- A system is said to be **time invariant** if, for any input $x(t)$ and any time t_1 , the response to the shifted input $x(t - t_1)$ is equal to $y(t - t_1)$ where $y(t)$ is the response to $x(t)$ with zero initial energy



- A system that is not time invariant is said to be **time varying** or **time variant**

Examples of Time Varying Systems

- Amplifier with Time-Varying Gain

$$y(t) = tx(t)$$

- First-Order System

$$\dot{y}(t) + a(t)y(t) = bx(t)$$

Basic System Properties: CT Linear Finite-Dimensional Systems

- If the N-th derivative of a CT system can be written in the form

$$y^{(N)}(t) = -\sum_{i=0}^{N-1} a_i(t) y^{(i)}(t) + \sum_{i=0}^M b_i(t) x^{(i)}(t)$$

then the system is both linear and finite dimensional

- To be time-invariant

$$a_i(t) = a_i \quad \text{and} \quad b_i(t) = b_i \quad \forall i \text{ and } t \in \mathbb{R}$$